National Skills University, Islamabad

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Course: Advanced Programming
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Program and Session: BS-CS , S26 - Morning (Section A) - 4th Semester
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Week 1: Lab

Code: Basic input and output in C#

```
/**
Name: Sumama Rahim
Reg# F24BSCS039
Date: 02/27/2026
**/

// using System;

// Week 01: Input (Strings and Numbers) and Output in C#
namespace Week01
{
 class Program
 {
   static void Main(string[] args)
   {
       // declaring variables
       String stdName, stdReg, stdClass;
       // Taking inputs
       Console.WriteLine("Enter Student Name: ");
       stdName = Console.ReadLine();
```

```
        Console.WriteLine("Enter Student Reg# ");
        stdReg = Console.ReadLine();
        Console.WriteLine("Enter Student Class");
        stdClass = Console.ReadLine();

        // reading numerical value
        double obtMarks, totalMarks, percentage;
        Console.WriteLine("Enter Total Marks");
        totalMarks = Convert.ToDouble(Console.ReadLine());
        Console.WriteLine("Enter Obtained Marks");
        obtMarks = Convert.ToDouble(Console.ReadLine());
        percentage = (obtMarks*100)/totalMarks;

        // Displaying Student Details
        Console.WriteLine(); // For blank line
        Console.WriteLine("*** Student Details ***");
        Console.WriteLine("Student Name: " + stdName
        + "\nStudent Reg# " + stdReg
        + "\nStudent Class: " + stdClass
        + "\nObtained Marks: " + obtMarks
        + "\nPercentage: " + percentage);

    }
  }
}
```

Output:

```
essaar@essaar-sys:~/Documents/Spring 26/Advanced Programming/Code/Week01$
Enter Student Name:
Sumama Rahim
Enter Student Reg#
F24BSCS039
Enter Student Class
4th Semester
Enter Total Marks
150
Enter Obtained Marks
141

*** Student Details ***
Student Name: Sumama Rahim
Student Reg# F24BSCS039
Student Class: 4th Semester
Obtained Marks: 141
Percentage: 94
```

Report on Program 1: Input and Output in C#

This report details the objective, structure, and execution of the first C# program presented, which focuses on handling input (both string and numerical) and displaying formatted output.

Program Objective

The primary objective of the first program was to practice:

1. **String Input:** Taking a student's name, registration number, and class as string input from the console.
2. **Numerical Input:** Accepting total marks and obtained marks as numerical input (specifically `double`).
3. **Calculation:** Computing the student's percentage based on the marks.
4. **Formatted Output:** Displaying all collected student details and the calculated percentage in a clear, multi-line format on the console.

Expected Output

The program is interactive, so the output depends on the values entered by the user. A representative execution scenario is documented below:

Test Case Scenario

Input Field	Data Type	Sample Input
Enter Student Name:	String	Sumama Rahim
Enter Student Reg#	String	F24BSCS039
Enter Student Class	String	BSCS
Enter Total Marks	Double	150
Enter Obtained Marks	Double	141

Expected Console Output

Based on the sample inputs, the console output should be:

```
Enter Student Name:
Sumama Rahim
Enter Student Reg#
F24BSCS039
Enter Student Class
BSCS
Enter Total Marks
150
Enter Obtained Marks
141

*** Student Details ***
Student Name: Sumama Rahim
Student Reg# F24BSCS039
Student Class: BSCS
Obtained Marks: 141
Percentage: 94
```

Code: Decision making in C#

```
// using System;

// Week 01: Input (Strings and Numbers) and Output in C#
namespace Week01
{
 class Decisions
 {
   static void Main(string[] args)
   {
     Console.WriteLine("\n\n***** Decisions in C# ******");

     // Decision making in C#
     // IF statement is same as it is in C/C++ or Java
     Console.WriteLine("Enter your fav number: ");
     int favNum = Convert.ToInt16(Console.ReadLine());

     if ((favNum%2 == 0) & (favNum != 0))
     {
       Console.WriteLine("You love Even numbers");
     } else if (favNum%2 == 1)
     {
       Console.WriteLine("You love Odd Numbers");
     } else if (favNum == 0)
     {
       Console.WriteLine("You love ZERO!");
     } else
     {
       Console.WriteLine("You don't love numbers!");
     }

     // Switch Statements in C#
     char gender;
     Console.WriteLine("Enter Gender (m/f): ");
     gender = Convert.ToChar(Console.ReadLine());
```

```
      switch (gender)
      {
        case 'm':
        Console.WriteLine("Male");
        break;
        case 'f':
        Console.WriteLine("Female");
        break;
        default:
        Console.WriteLine("Enter only m or f.");
        break;
      }
    }
  }
}
```

Output:

Test Case 1

```
(.venv) essaar@essaar-sys:~/Documents/Spring 26/Advanced Programming/Code/Week01$ dotnet run


***** Decisions in C# ******
Enter your fav number:
7
You love Odd Numbers
Enter Gender (m/f):
m
Male
(.venv) essaar@essaar-sys:~/Documents/Spring 26/Advanced Programming/Code/Week01$
```

Test Case 2

```
^[OA^C(.venv) essaar@essaar-sys:~/Documents/Spring 26/Advanced Programming/Code/Week01$ dotne
run

***** Decisions in C# ******
Enter your fav number:
8
You love Even numbers
Enter Gender (m/f):
f
Female
(.venv) essaar@essaar-sys:~/Documents/Spring 26/Advanced Programming/Code/Week01$
```

Test Case 3:

```
(.venv) essaar@essaar-sys:~/Documents/Spring 26/Advanced Programming/Code/Week01$ dotnet run


***** Decisions in C# ******
Enter your fav number:
0
You love ZERO!
Enter Gender (m/f):
m
Male
(.venv) essaar@essaar-sys:~/Documents/Spring 26/Advanced Programming/Code/Week01$
```

Report on Test Cases for Decision Making in C# Code

The following report outlines the execution and expected output for three test cases against the C# code block titled "Code: Decision making in C#". This code primarily tests `if-else if-else` structures for numerical input (identifying even, odd, or zero) and `switch` statements for character input (identifying gender).

Detailed Test Case Results

Test Case	Input (If else if)	Input (Switch)	Expected Output (Favorite Number Logic)	Expected Output (Gender Switch)
Test Case 1	10	m	You love Even numbers	Male
Test Case 2	5	f	You love Odd Numbers	Female
Test Case 3	0	x	You love ZERO!	Enter only m or f.

Test Case 1: Even Number and Male Gender

- **Input:**
  - Enter your fav number: `10`
  - Enter Gender (m/f): `m`
- **Expected Console Output:**

You love Even numbers
Male

Test Case 2: Odd Number and Female Gender

- **Input:**
    - Enter your fav number: `5`
    - Enter Gender (m/f): `f`
- **Expected Console Output:**

You love Odd Numbers
Female

Test Case 3: Zero and Invalid Gender

- **Input:**
    - Enter your fav number: `0`
    - Enter Gender (m/f): `x`
- **Expected Console Output:**

You love ZERO!
Enter only m or f.

THE END